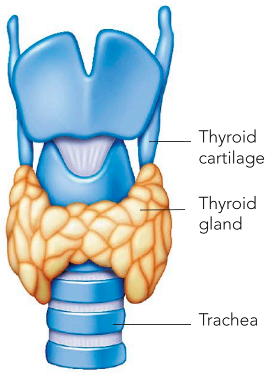
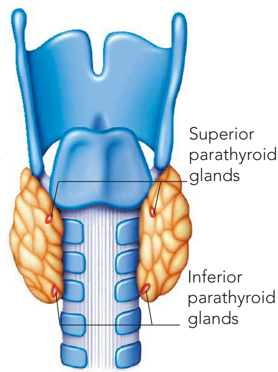
THYROID AND PARATHYROID GLANDS

The thyroid is located in the front of the neck, and has four tiny parathyroid glands embedded at the back. The hormones it produces have wide-ranging effects on body chemistry, including the maintenance of body weight, the rate of energy use from blood glucose, and heart rate. Unlike other glands, the thyroid can store its hormones. The parathyroids make parathormone (PTH), which increases the levels of calcium in the blood. PTH acts on bones to release their stored calcium, on the intestines to increase calcium absorption, and on the kidneys to prevent calcium loss.



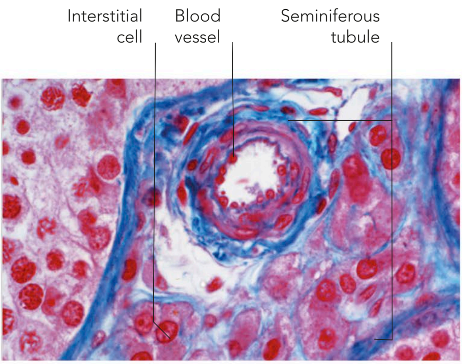
THYROID
The thyroid wraps round the upper windpipe (trachea). It produces two hormones that regulate the body's metabolism: thyroxine (T_4) and triiodothyronine (T_3).



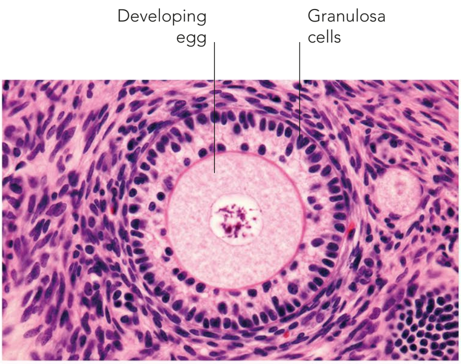
PARATHYROIDS
The small parathyroid glands are set into the rear corners of the thyroid's lobes, at the back of the trachea. There are usually four, but their number and exact locations vary.

SEX GLANDS AND HORMONES

The main sex glands are the ovaries and testes. The hormones they produce stimulate the production of eggs and sperm respectively, and influence a developing embryo's sex. Until puberty, levels of sex hormones remain low. Then, in males, the testes increase their output of androgens (male sex hormones), such as testosterone. In females, the ovaries produce more oestrogens and progesterone.



TESTOSTERONE PRODUCERS
The cells shown in pink in this microscopic image of the testis secrete testosterone. They are found in the connective tissue between seminiferous tubules.



OESTROGEN PRODUCERS
This microscope picture shows a developing egg surrounded by a ring of granulosa cells. These cells secrete oestrogens.